# Practice workbook · Network types: LAN, WAN, PAN and why we network

Reasons for networking computers; naming or contrasting LAN, WAN and PAN from a described scenario; client-server versus peer-to-peer usage models.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-01

# 1. Question 1 · part a · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

**1** There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

**(2)**

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

**(1)**

.....

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## 1. Question 1 · part a · 2 marks · Working and answer

Working and answer

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# 2. Question 3 · part a.i · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

**(1)**

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

**(1)**

.....

.....

(b) Identify the measurement used for network data speeds.

**(1)**

☒ **A** Megabits per second

☒ **B** Megapixels per second

☒ **C** Mebicycles per second

☒ **D** Mebihertz per second

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## 2. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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# 3. Question 3 · part a.ii · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part a.ii only.

**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

**(1)**

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

**(1)**

.....

.....

(b) Identify the measurement used for network data speeds.

**(1)**

☒ **A** Megabits per second

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☒ **C** Mebicycles per second

☒ **D** Mebihertz per second

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## 3. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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## 4. Question 1 · part a · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

**1** Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

**(2)**

1 .....

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2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

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## 4. Question 1 · part a · 2 marks · Working and answer

Working and answer

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# 5. Question 2 · part a · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part a only.

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**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

.....

.....

(b) State the purpose of network protocols.

**(1)**

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

**(1)**

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

P 6 6 4 8 9 A 0 5 2 0

## 5. Question 2 · part a · 1 mark · Working and answer

Working and answer

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## 6. Question 3 · part a · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

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**3** Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

**(2)**

| Network type | Tick (✓) | Usage model | Tick (✓) |
|---|---|---|---|
| Local Area Network (LAN) | | Client-Server | |
| Wide Area Network (WAN) | | Peer-to-peer | |
| Personal Area Network (PAN) | | | |

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

**(4)**

| | Layer name | Function |
|---|---|---|
| 1 | | |
| 2 | | |

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## 6. Question 3 · part a · 2 marks · Working and answer

Working and answer

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# 7. Question 5 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

**5** Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

**(2)**

(ii) Give **one** reason why a patient may not want to use this online service.

**(1)**

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

**(1)**

(ii) Name the type of network used to access the server from the Stockton building.

**(1)**

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P 7 2 5 3 8 A 0 1 6 2 0

## 7. Question 5 · part b.i · 1 mark · Working and answer

Working and answer

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# 8. Question 5 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

**5** Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

**(2)**

(ii) Give **one** reason why a patient may not want to use this online service.

**(1)**

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

**(1)**

(ii) Name the type of network used to access the server from the Stockton building.

**(1)**

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## 8. Question 5 · part b.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.ii only.

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(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

| Description | RAM | ROM |
|---|---|---|
| Stores the boot up sequence | | |
| The contents are lost when the laptop is shut down | | |

**(Total for Question 5 = 12 marks)**

## 8. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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## 9. Question 6 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

**6** Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

**Figure 3** shows a network topology.

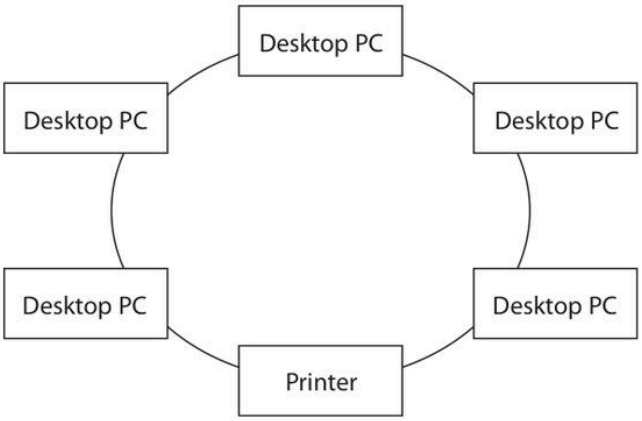



**Figure 3**

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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P 7 2 5 3 8 A 0 1 8 2 0

## 9. Question 6 · part b.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

**(1)**

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

**(2)**

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

**(1)**

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

**(1)**

## 9. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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## 10. Question 1 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

**1** People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

**(1)**

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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## 10. Question 1 · part a · 2 marks · Working and answer

Working and answer

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